Rajarambapu Institute of Technology, Rajaramnagar

Quantum 2k26

PharmaBharat — Verified B2B Pharmaceutical Supply, Redistribution & Retail Intelligence Platform

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Background

India dispenses more medicine than any nation on earth through roughly **900,000 independent pharmacies** — and almost none of them can see the single number that decides whether stock becomes revenue or waste: **the batch expiry date**. It is printed on the carton, copied onto a paper bill, and then it disappears.

PharmaBharat rebuilds that missing layer. It is one verified network on which retailers, wholesale distributors and pharmaceutical manufacturers trade **at the level of the batch** — and on which every medical store simultaneously runs its entire business: unit-level stock, GST counter billing, customers, credit and statutory compliance, all on the same data.

Manufacturer *batch created* → **Distributor** *batch moved* → **Pharmacy** *batch dispensed* → **Patient** *batch on the bill*

One unbroken chain of custody — godown to the single tablet cut at the counter.

Problem Statement

Pharmaceutical retail loses money in three places at once — and all three descend from the same blind spot.

- **On the shelf.** Medicines expire unseen. An estimated **3–5% of stock value** is written off to expiry across the distribution chain, a loss carried almost entirely by the smallest players.
- **At the point of purchase.** The same molecule carries different prices from different distributors, discoverable only by telephone. Most shops never compare — they reorder from habit.
- **In compliance.** Schedule H/H1 registers are handwritten, drug licences lapse unnoticed, and a batch recall still travels by phone call — if it travels at all.

Nobody can see the batch. Everything else follows from that.

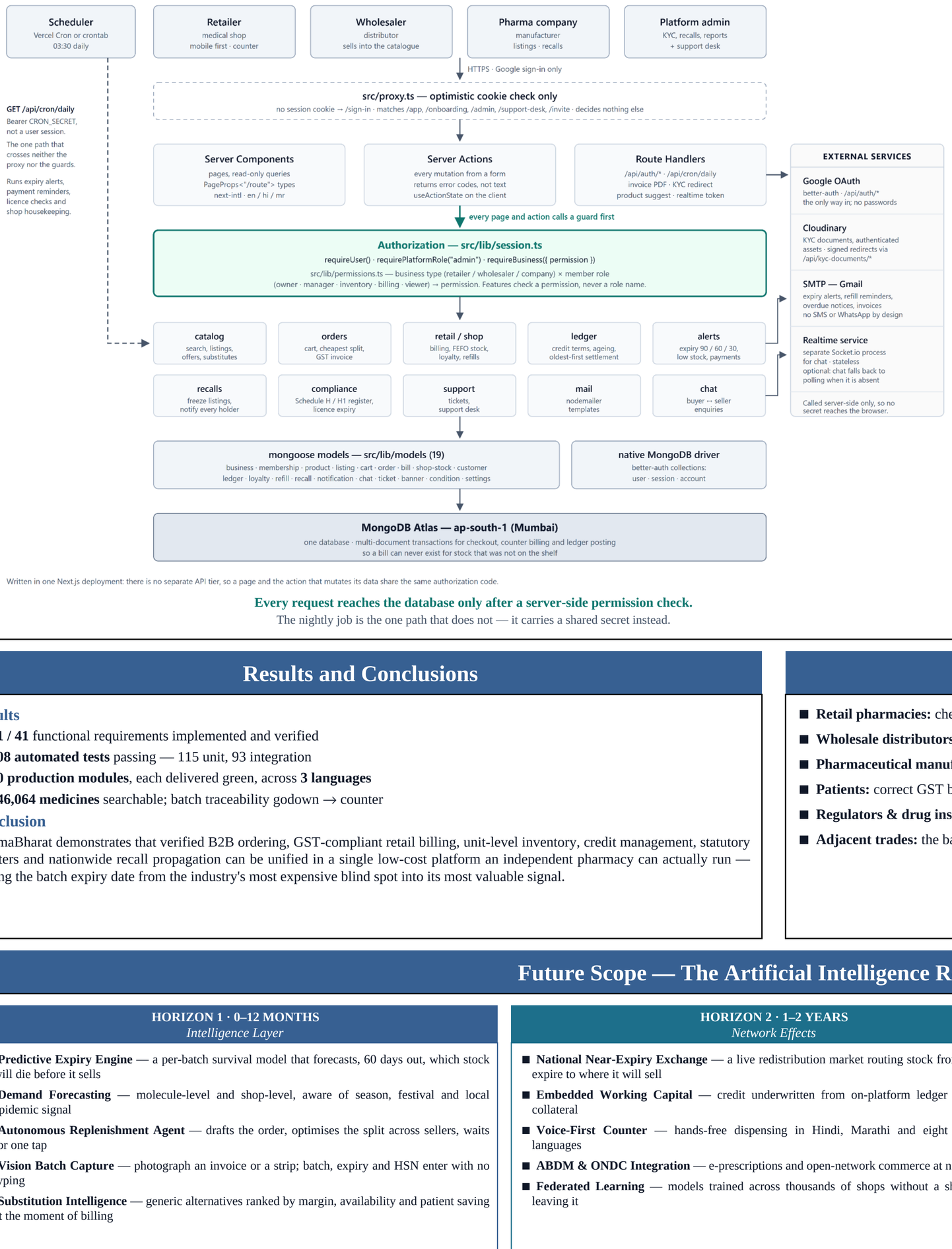
Objectives

- Verify **every** trading entity through KYC — drug licence, GST position and PAN — before a single transaction.
- Make the **batch and its expiry date** a first-class, visible object across the whole supply chain.
- Let a pharmacy compare every verified seller on **price, margin and expiry** before committing a rupee.
- Run the complete retail counter — **unit-level stock, GST billing, loyalty, credit** — on the same data.
- Convert expiry from an invisible loss into a **measured, escalating 90/60/30-day signal**.
- Make compliance automatic: **digital H/H1 register, licence enforcement, instant recall propagation**.

Methodology

1. **Requirement Engineering** — 41 functional requirements, SRS
2. **Domain & System Design** — modular monolith, permission matrix
3. **Iterative Development** — 10 modules, built in dependency order
4. **Security & Compliance** — RBAC, ACID transactions, KYC
5. **Verification** — 208 automated tests, end-to-end typing
6. **Evaluation & Pilot Readiness** — seeded demo shop, deployment

Project Diagram — System Architecture



Every request reaches the database only after a server-side permission check.
The nightly job is the one path that does not — it carries a shared secret instead.

Technical Specifications

- **Frontend:** Next.js 16 App Router, React 19, TypeScript, Tailwind CSS
- **Backend:** React Server Components + Server Actions — no separate API tier, so a page and the action that mutates it share one authorization path
- **Database:** MongoDB Atlas (ap-south-1 Mumbai), Mongoose ODM, 19 domain models
- **Integrity:** multi-document **ACID transactions** on checkout, counter billing and ledger posting
- **Identity:** better-auth with Google OAuth — no passwords ever stored
- **Authorization:** permission matrix of business type × member role, enforced server-side on **every** action; the edge proxy authorises nothing
- **Dispensing:** FEFO engine — the batch expiring first is always offered first
- **Taxation:** CGST/SGST/IGST engine, MRP-inclusive reverse computation, three GST positions (registered, composition, unregistered)
- **Compliance:** Schedule H/H1 digital register, licence-expiry enforcement, platform-wide recall freeze
- **Localisation:** English, हिन्दी, मराठी — key parity enforced by automated test
- **Documents:** Cloudinary authenticated assets served only via signed redirects
- **Catalogue:** 246,064-medicine national master with fuzzy and salt-level search

Results and Conclusions

Results

- **41 / 41** functional requirements implemented and verified
- **208 automated tests** passing — 115 unit, 93 integration
- **10 production modules**, each delivered green, across **3 languages**
- **246,064 medicines** searchable; batch traceability godown → counter

Conclusion

PharmaBharat demonstrates that verified B2B ordering, GST-compliant retail billing, unit-level inventory, credit management, statutory registers and nationwide recall propagation can be unified in a single low-cost platform an independent pharmacy can actually run — turning the batch expiry date from the industry's most expensive blind spot into its most valuable signal.

Applications

- **Retail pharmacies:** cheaper verified buying, plus a complete digital counter that replaces the register
- **Wholesale distributors:** verified demand, faster movement of near-expiry stock, fewer credit disputes
- **Pharmaceutical manufacturers:** true sell-through visibility and instant, provable recall reach
- **Patients:** correct GST bills, refill reminders, substitution savings and recall safety
- **Regulators & drug inspectors:** machine-readable H/H1 registers and live licence status — an inspection becomes a download
- **Adjacent trades:** the batch-and-expiry core extends directly to agro-chemical, veterinary and perishable FMCG distribution

Future Scope — The Artificial Intelligence Roadmap

HORIZON 1 · 0–12 MONTHS
Intelligence Layer

- **Predictive Expiry Engine** — a per-batch survival model that forecasts, 60 days out, which stock will die before it sells
- **Demand Forecasting** — molecule-level and shop-level, aware of season, festival and local epidemic signal
- **Autonomous Replenishment Agent** — drafts the order, optimises the split across sellers, waits for one tap
- **Vision Batch Capture** — photograph an invoice or a strip; batch, expiry and HSN enter with no typing
- **Substitution Intelligence** — generic alternatives ranked by margin, availability and patient saving at the moment of billing

HORIZON 2 · 1–2 YEARS
Network Effects

- **National Near-Expiry Exchange** — a live redistribution market routing stock from where it will expire to where it will sell
- **Embedded Working Capital** — credit underwritten from on-platform ledger behaviour, not collateral
- **Voice-First Counter** — hands-free dispensing in Hindi, Marathi and eight further Indian languages
- **ABDM & ONDC Integration** — e-prescriptions and open-network commerce at national scale
- **Federated Learning** — models trained across thousands of shops without a shop's data ever leaving it

HORIZON 3 · 2–5 YEARS
Category-Defining Infrastructure

- **Spurious-Drug Detection** — graph anomaly detection over batch provenance to surface counterfeit flows
- **Blockchain-Anchored Provenance** — immutable manufacturer-to-patient chain of custody
- **IoT Cold-Chain Digital Twin** — live temperature telemetry bound to the batch record itself
- **Clinical Adherence Engine** — chronic-patient refill prediction and timely intervention
- **National Pharmaco-Signal Grid** — anonymised real-time dispensing as public-health early warning

Every model above trains on data the platform already captures. The hard part — getting every batch and every date into a system at all — is built.